The finite moving graphs in the triality examples

Human Compression project: L017 finite comparison

We compare every example on page 16 of Leemans–Stokes–Tranchida, *On trialities and their absolute geometries*, 21 March 2024. The source supplies no generator matrices or executable program for these examples. The literal source claims and the present explicit witnesses must therefore be distinguished. The original constructions belong to the three source authors; the complete marking reduction, independent implementation and corrections below are this recovery's contribution. This author submission still requires distinct review and whole-original acceptance.

1 A complete and small matrix boundary

Put $Q = q^3$, $F = \mathbb{F}_Q$, $k = \mathbb{F}_q$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. We use the actual automorphism $\sigma : X \mapsto X^{(q)}$. The paid uniform proof in *Absolute flags and their moving lines* establishes full completion in both characteristics and the exact fixed absolute. Projective-linear descent conjugates any source marking's triality to σ or σ^{-1}. Replacing it by its inverse does not change either absolute. Its fixed middle involution can be conjugated to the projective image of J over k. In odd characteristic its determinant-one lift is itself fixed by σ: a negative lift would remain negative after three applications. The rational canonical form of this trace-zero matrix gives the asserted conjugation. In characteristic two all nonidentity unipotents are conjugate over the perfect field k.

Choose the unique sign of the lift R of the first endpoint for which

$$\det R = 1, \quad \operatorname{tr} R = 0, \qquad R^{\sigma^2} = RR^{\sigma}. \tag{1}$$

In odd characteristic the two lifts change the sign of the left side and not of the right side, so exactly one works. In characteristic two there is no sign choice. Thus every source marking occurs among the solutions of (1), with $a = \overline{R}$, $b = \overline{J}$, $c = \overline{R^{\sigma}}$.

We retain in full the accepted three-page L023 matrix-equation proof. It proves that the odd-characteristic solutions form one free $\mathrm{PGL}_2(k)$ conjugacy orbit, of size $q(q^2-1)$, and that the nonscalar even solutions consist of a k-rational fixed point and a nonzero trace-zero parameter, of size $(q+1)(q^2-1)$. These are written exhaustion theorems, not inferred from a successful search. The present program checks its literal odd seed against (1), enumerates that complete orbit, and enumerates all the even parameters. It then removes repetitions under the projective centralizer of J over k and coordinate Frobenius. Both operations preserve the resulting moving graph. Consequently this reduction can establish nonexistence of a claimed graph as well as produce witnesses. It is used exhaustively for $q = 7$; the other table entries below need only their displayed witnesses.

The five fields are given in the program by these explicit moduli; z is the residue of the indeterminate:

q	p	modulus over $\mathbb{F}_p$
2	2	$z^3 + z + 1$
4	2	$z^6 + z^4 + z^3 + z + 1$
5	5	$z^3 + 3z + 3$
7	7	$z^3 + 6z^2 + 4$
9	3	$z^6 + 2z^4 + z^2 + 2z + 2$

The program checks that successive powers of z visit every nonzero residue once and return to one. This also proves that every nonzero residue is a unit, hence that the quotient ring is a field. No irreducibility or primitive-polynomial database is an unchecked premise.

2 Exact construction and finite trust boundary

Let $H_0 = \langle R, J \rangle$, $V = \langle R, R^\sigma \rangle$ in the determinant-one matrix group, including its scalar center, and let $H = \mathrm{SL}_2(k)$. The uniform fixed-absolute theorem proves that all absolute right cosets of H_0 are among

$$H_0 h, \quad H_0 R^\sigma h \qquad (h \in H). \tag{2}$$

The program forms each whole coset, identifies duplicates by its actual matrices, and enumerates all original edges at these vertices. An edge represented by Vx has other endpoint $H_0 R^\sigma x$ and is fixed exactly when $x^\sigma x^{-1} \in V$. It is retained as moving precisely when its two endpoints belong to (2) and it is not fixed. Distinct edge cosets are checked against their endpoint pairs, so parallel edges cannot silently disappear. Isolated vertices are then removed exactly as specified in the source. The matrices, whole cosets, adjacency lists and edge-fixedness tests are exact finite-field operations.

For every displayed witness, and all seven $q = 7$ representatives, generation of $\mathrm{PSL}_2(Q)$ can also be read from the marked period m. Usually $m = (Q-1)/\gcd(2, Q-1)$ or $(Q+1)/\gcd(2, Q-1)$, so the two-generator dihedral is a full torus normalizer. The retained Dickson classification makes it maximal here. The generated group is triality-stable and contains a Klein four subgroup cycled by the triality. It cannot be dihedral: an odd rotation subgroup admits no Klein four, whereas an even one makes each Klein four contain the characteristic central involution, contradicting that cycle. The single remaining $q = 7$ period is 86. Its prime factor 43 excludes the Borel, subfield and exceptional maximal subgroups of $\mathrm{PSL}_2(343)$ by their orders; the remaining possible dihedral containment is excluded by the same central-involution argument. This is a named use of Dickson's classification, not a new subgroup classification.

Finally these generating markings have no marked duality. If an automorphism fixed b and interchanged a, c, it and the displayed triality would induce the faithful S_3 on the three nonidentity elements of V. Since the markings generate, the corresponding relations hold as actual automorphisms of G. Their commutator is the noninner triality or its inverse. This is impossible in the retained abelian outer automorphism group of $\mathrm{PSL}_2(Q)$. Thus every displayed witness meets source condition (5), not merely the involution and generation conditions.

The full required programs are `finite_graphs.cpp` and `graph_certificate.py`; the literal automorphism generators are required data. The first produces the complete relevant marking lists and coset graphs. The second checks adjacency, distances, components and full automorphism orders. These are finite mathematical premises. Their successful termination and ordinary exact arithmetic are trusted; no kernel-checked exhaustion or formally verified compiler is claimed. A separate GAP field, group and coset reconstruction agrees with all selected graphs and all seven $q = 7$ cases; it is a cross-check, not a substitute for the written exhaustion.

3 Full automorphism certificates

For a graph, partition vertices by any stated intrinsic invariants, then repeatedly refine each cell by the numbers of neighbors in every current cell. Every automorphism preserves this partition. Individualize a vertex in a nonsingleton cell of size b and refine again. The stabilizer index is at most b. If this process ends in singletons, the product of these cell sizes is an upper bound on the full automorphism group. This argument does not assume that a refinement cell is an actual orbit.

The certificate verifies literal permutation generators against every edge and closes the small generated groups exactly. Matching upper and lower bounds prove fullness. For the connected

cubic 90-vertex graph, the initial invariant is the number of five-cycles through a vertex: its cells have sizes 30 and 60. All other initial partitions are indiscrete. The certificate's stabilizer-index bounds are

graph	refinement factors	$\lvert \mathrm{Aut} \rvert$
6-vertex prism	$6 \cdot 2$	12
90-vertex fixed–moving union	$30 \cdot 2 \cdot 2$	120
30-vertex moving	$30 \cdot 4 \cdot 2$	240
60-vertex moving	$60 \cdot 2$	120
168-vertex moving	$168 \cdot 2^{29}$	90 194 313 216
180-vertex moving	$180 \cdot 2 \cdot 4$	1440

For the large order no enormous permutation closure is required. Every vertex lies in a unique triangle; the 56 triangles occur in 28 intrinsic pairs with identical neighborhoods in their triangle quotient. Each pair supports a literal six-point interchange fixing every other vertex. The certificate verifies all 28 interchanges. Their disjoint supports give an elementary abelian group of order 2^{28} in the kernel of the action on the 28 six-point blocks. The supplied graph automorphisms induce a group of order 336 on those blocks, checked by direct closure. Thus the full group has order at least $2^{28} \cdot 336$, exactly the refinement upper bound. In particular it is not isomorphic to $\mathrm{PSL}_2(7)$.

For each claimed $C_2 \times A_5$ identification the certificate finds a center of order two and a derived subgroup of order 60, with trivial intersection. The derived subgroup acts faithfully on its five Klein four subgroups, identifying it with A_5. Their orders prove the direct product. For the 240-group it additionally finds an involution extending the derived subgroup to a subgroup of order 120, disjoint from the center and acting faithfully on the same five Klein groups. This subgroup is S_5, proving the direct product $C_2 \times S_5$. The finite group identifications therefore do not rest on a database's abstract-name output.

4 Every reported example

The following are exact outputs for the indicated current matrix records. Record numbers are our deterministic finite-field ordering, not invented source labels. Diameters in this table refer to ordinary graph distance.

q	record	v	e	moving graph	g	d
2	0	6	9	triangular prism	3	2
4	0	90	75	$15K_2 \sqcup 12C_5$	5	∞
5	3	30	60	connected, degree 4	5	3
5	4	60	120	connected, degree 4	3	8
7	3	168	336	connected, degree 4	3	9
9	0	180	360	connected, degree 4	5	6

The 60-vertex graph is vertex-transitive with full group $C_2 \times A_5$; the 30-vertex graph is arc-transitive with full group $C_2 \times S_5$. The latter's incidence point and line eccentricities are 7 and 8, and its incidence gonality is 5, exactly the source diagram. For the 180-vertex graph both incidence eccentricities are 13 and the gonality is 5, again exactly the source diagram. The first record has nine absolute vertices before deleting three isolated moving vertices. Its fixed graph is three two-edge paths, and its moving graph is the source's triangular prism. There is one generating marking class under semilinear conjugacy. This is the precise conjugacy convention verified here; it is not a claim of one ordered inner-conjugacy class.

The $q = 4$ paragraph conflates two graphs. Its moving graph really is $15K_2 \sqcup 12C_5$, with degrees one and two, and 75 edges. Its full automorphism group has order

$$2^{15}\, 15!\, 10^{12}\, 12!,$$

by permuting and independently acting on its connected components. It is disconnected, so has no finite diameter. On the same 90 absolute vertices, *adding the 60 fixed edges* gives a connected cubic graph with 135 edges, girth five, diameter seven and full group $C_2 \times A_5$. Thus the later properties in the printed paragraph are recovered for the fixed–moving union, not falsely attributed to the moving graph.

For $q = 7$, all 336 normalized solutions of (1) reduce to seven classes under the certified graph-preserving operations. Each generates the full group. After removing isolated vertices their complete list is

$$84K_2, \quad \varnothing, \quad 42C_4, \quad \text{the connected 168-vertex graph}, \quad 21C_{12}, \quad 42D, \quad 84P_3, \tag{3}$$

where D is the six-vertex tree obtained by joining the centers of two two-leaf stars. In particular the asserted 84-vertex degree-four example does not occur for any marking satisfying the source hypotheses. This is an exhaustive negative conclusion from (1)–(3), not a failure to find a witness in a sample. The 168-vertex degree-four graph has the stated connectedness, girth and diameter, but its asserted full automorphism group is corrected by the exact certificate above. These two source claims are therefore explicitly refuted in their literal stated scope.

Current contract. The complete classical moving theorem and the fixed-absolute theorem are recovered with explicit retained inputs and full written proofs. All seven reported finite examples are accounted for, including the two $q = 7$ corrections and the $q = 4$ graph distinction. The definition of the absolute uses minimal invariant flags, and the finite algorithm uses an actual semilinear automorphism. This recovery includes corrections; it does not assert that every original statement is true. Type-II trialities mentioned as future work are not solved here. No new conjecture, formalization or compression factor is claimed.